

MARISSA CLARK

Chicago, IL | marissadclark@gmail.com | linkedin.com/in/marissa-d-clark

EXPERIENCE

Senior Data Scientist | United Airlines | 2024–Present

Translating audience targeting and credit card offer challenges into modeling and optimization solutions, working with marketing and growth teams to support strategic decisions across the 100M+ member MileagePlus program.

- Designed and implemented a machine learning ranking model to automate audience selection and content prioritization for campaigns reaching **12M+ MileagePlus members**, validated via A/B testing (RCTs); replaced manual targeting workflows, **increased clicks per recipient by 9%**, and **reduced unsubscribe rates by up to 20%**.
- Built a prescriptive optimization model in Python using **PuLP** to determine optimal co-branded credit card offer assignments under complex business constraints. The model replaced a vendor solution and contributed to a **20% lift in direct mail conversion rates**, enabling approximately **\$8M in projected incremental revenue**.
- Led the design of scalable analytics workflows to triangulate first-party and third-party (3P) data using **AWS Clean Rooms, Habu, and LiveRamp**, evaluating data quality and audience overlap across partner sources. Delivered insights that informed **\$10M+ in partnership agreements**.
- Hired and mentored a summer data science intern through his first end-to-end ML project, guiding an XGBoost model predicting non-member credit card sign-up likelihood to inform direct mail and email campaign targeting; set clear objectives and coached his development through a mid-summer check-in, and he **was recommended by leadership to return for a second internship**.

Data Science Strategy Lead | Kimberly-Clark | 2023–2024

Partnered with marketing and commercial teams to translate customer acquisition and demand planning challenges into predictive modeling, forecasting, and 3P data strategy solutions supporting brand growth and trade promotion efficiency.

- Evaluated the strategic value and predictive signal of a **3P consumer data provider** across a **1.7M+ household footprint**, assessing data quality and reliability to inform a **multi-million-dollar data acquisition contract** and long-term data strategy.
- Developed a predictive model using Scikit-Learn to identify **first-time Huggies purchasers** prior to their initial transaction, enabling proactive marketing outreach and strengthening early-stage brand loyalty.
- Developed a trade promotion optimization algorithm to prioritize promotional timing for **5,700+ products**; drove an estimated **\$3.3M in forecasted incremental sales**.
- Delivered hands-on training on regression-based forecasting in Python and Power BI to sales and commercial stakeholders; improved demand planning alignment and boosted sales team adoption by **40%**.

Data Scientist | Home Chef | 2021–2022

Applied predictive modeling, attribution analysis, and statistical methods to improve marketing efficiency, customer acquisition strategy, and operational forecasting.

- Analyzed the impact of **TV advertising** on customer acquisition using SQL, Python, and XGBoost, applying Fast Fourier Transform analysis to separate daily and hourly demand patterns and isolate incremental lift; insights contributed to budget decisions and **\$500K in annual savings**.
- Built a **customer lifetime value model** that guided budget allocation and reshaped lifecycle marketing efforts.
- Improved **meal-kit sales forecasting** using historical sales and recipe data in **Snowflake**, reducing food waste and saving approximately **\$750K annually**.
- Developed an **NLP pipeline** to classify customer feedback and surface emerging trends, enabling faster response to shifts in customer sentiment.

PhD Researcher | Dartmouth College | Computational Social Neuroscience Lab | 2018–2021

Investigated how the brain represents unique emotional experiences, using autobiographical video stimuli and a probabilistic Shared Response Model to uncover both shared neural patterns and significant individual variability in the brain's default mode network.

- Managed a **\$3.5M academic research grant**, leading data collection and advanced statistical modeling in Python and R, including shared response modeling, factor analysis, and PCA, to understand emotional processing in the brain.
- Presented research findings to technical and non-technical audiences at national conferences; contributed to study design, analysis planning, and interpretation.

EDUCATION

Dartmouth College | PhD Student in Psychological and Brain Sciences (coursework completed) | 2018–2021

UCLA | Bachelor of Science in Cognitive Science, Minor in Neuroscience | 2015

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, R, Bash

Modeling & Statistical Analysis: Machine Learning, Predictive Modeling, Causal Inference (A/B Testing, RCTs), Time Series, Optimization (PuLP), Forecasting, Hypothesis Testing, Linear & Logistic Regression, Multilevel Modeling, NLP

Data & Platforms: AWS, Clean Rooms (Habu, LiveRamp), Snowflake, SageMaker, Airflow, dbt, Looker, Jupyter, Git, Databricks